

Conventions and Cross-Validation Notes: Momentum-Space Grouped MPS Ansatz for the 1D t - V Model

grouped.ansatz project notes

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1 The model

We study the 1D spinless-fermion t - V chain at half filling ($N = L/2$ particles on L sites), $t = 1$ throughout.

1.1 Real space

$$H = -t \sum_{i=1}^{L-1} \left(c_i^\dagger c_{i+1} + \text{h.c.} \right) - t \cos \theta \left(c_1^\dagger c_L + \text{h.c.} \right) + V \sum_{i=1}^{L-1} n_i n_{i+1} + V n_1 n_L, \quad (1)$$

with $n_i = c_i^\dagger c_i$ and twist angle $\theta = \pi$ (antiperiodic hopping boundary condition: the wraparound bond $c_1^\dagger c_L$ picks up a sign flip, $-t \cos \pi = +t$). The interaction is a *plain* density–density term, coefficient V on every bond including the wraparound — density operators commute, so this term needs no boundary twist of its own (only the hopping does).

1.2 Momentum space

Fourier transform with half-odd-integer momentum quantization, matching the antiperiodic real-space boundary:

$$k_i = -\pi + \frac{\pi(2i-1)}{L}, \quad i = 1, \dots, L, \quad (2)$$

giving L points evenly spaced by $2\pi/L$, symmetric about $k = 0$, with no point exactly at $k = 0$ or $k = \pi$. The Hamiltonian becomes

$$H = \sum_{i=1}^L \varepsilon(k_i) n_{k_i} + \frac{1}{L} \sum_{i,j=1}^L \sum_{q=0}^{L-1} V \cos\left(\frac{2\pi q}{L}\right) c_{k_i+q}^\dagger c_{k_j-q}^\dagger c_{k_j} c_{k_i}, \quad (3)$$

with $\varepsilon(k) = -2t \cos k$, and q an integer shift of the *mode index* (so $k_i + q$ means “the mode q steps away from mode i ,” a genuine momentum shift of $2\pi q/L$ since the modes in Eq. (2) are evenly spaced).

1.3 Why the vertex is $V \cos q$, not $2V \cos q$

This is worth deriving explicitly since an earlier version of this codebase used a $2V$ vertex by mistake (see §4). Write $c_i = L^{-1/2} \sum_k e^{ikx_i} c_k$ with $x_i = i$. Then

$$\sum_{i=1}^L n_i n_{i+1} = \frac{1}{L^2} \sum_i \sum_{p_1, p'_1, p_2, p'_2} e^{i(p'_1 - p_1)x_i} e^{i(p'_2 - p_2)x_{i+1}} c_{p_1}^\dagger c_{p'_1} c_{p_2}^\dagger c_{p'_2}. \quad (4)$$

Summing over i enforces momentum conservation, $p'_1 + p'_2 = p_1 + p_2 \pmod{2\pi}$. Writing $q \equiv p'_1 - p_1$ (so $p'_2 - p_2 = -q$) and using $x_{i+1} = x_i + 1$:

$$\sum_i n_i n_{i+1} = \frac{1}{L} \sum_{p_1, p_2, q} e^{-iq} c_{p_1}^\dagger c_{p_1+q} c_{p_2}^\dagger c_{p_2-q} \equiv \frac{1}{L} \sum_{p_1, p_2, q} e^{-iq} F(p_1, p_2, q). \quad (5)$$

One shows $F(p_2, p_1, -q) = F(p_1, p_2, q)$ as an operator identity (moving a pair of fermion operators past another pair costs $(-1)^{2 \times 2} = +1$), so after summing over the dummy indices p_1, p_2 , $F(q) \equiv \sum_{p_1, p_2} F(p_1, p_2, q)$ satisfies $F(q) = F(-q)$. Splitting $e^{-iq} = \cos q - i \sin q$, the $\sin q$ part is odd under $q \rightarrow -q$ while $F(q)$ is even, so it cancels exactly in the sum over the full Brillouin zone $q = 0, \dots, L-1$ — leaving, with *no extra factor of 2*,

$$\sum_i n_i n_{i+1} = \frac{1}{L} \sum_{p_1, p_2, q} \cos q c_{p_1}^\dagger c_{p_1+q} c_{p_2}^\dagger c_{p_2-q}. \quad (6)$$

Hence $V \sum_i n_i n_{i+1}$ maps to exactly the vertex $V \cos q$ used in Eq. (3) — coefficient 1, confirming the real-space bond coefficient and the momentum-space vertex coefficient must be numerically identical (both called V) for the two representations to describe the same Hamiltonian.

1.4 XXZ mapping and the phase transition

Via Jordan–Wigner, Eq. (1) maps to the XXZ chain $H_{XXZ} = -t \sum_i (S_i^+ S_{i+1}^- + \text{h.c.}) + V \sum_i S_i^z S_{i+1}^z$ (dropping a boundary/linear term that vanishes at half filling), i.e. anisotropy

$$\Delta = \frac{V}{2t}. \quad (7)$$

The well-known gapless Luttinger-liquid ($\Delta < 1$) to gapped charge-density-wave ($\Delta > 1$) transition sits at $\Delta = 1$, i.e.

$$\boxed{V_c = 2t}. \quad (8)$$

With the corrected single- V convention above, this is exactly the “textbook” value; under the old (erroneous) $2V$ convention the same physical transition sat at $V_{\text{target}} = t$ instead, since the real bond coefficient was $2V_{\text{target}}$.

2 The grouped ($k, k+\pi$) ansatz

The variational ansatz used throughout this project (`grouped_mps.jl`) groups each momentum mode k with its partner $k + \pi$ into a single dimension-4 MPS supersite, rather than treating all L modes as separate dimension-2 sites. This choice is directly motivated by the structure of Eq. (3): the interaction vertex $V \cos q$ is maximal in magnitude at $q = 0$ (forward scattering) and $q = \pi$ (backscattering/umklapp, momentum transfer exactly π), and suppressed at intermediate q . The $q = \pi$ channel — pairs of modes separated by exactly π — is therefore the single strongest correlation channel in the model, independently confirmed for the (spinful, on-site- U) Hubbard model in momentum space by Ehlers, Sólyom, Legeza & Noack [Phys. Rev. B **92**, 235116 (2015)], who find empirically that reordering the DMRG chain to place $k, k + \pi$ pairs adjacent is the single most effective entanglement-reducing move. The grouped ansatz used here takes this one step further: instead of merely placing $k, k + \pi$ *adjacent* in the MPS chain (paying for their correlation via a truncated bond), it *fuses* them into one local tensor, absorbing that correlation entirely into the local physical dimension and paying nothing in bond dimension for it.

3 Numerical verification

All checks below were run after fixing the interaction coefficient consistently in every file that constructs H : the live VMC sampler (`get_E_loc` in `grouped_mps.jl`), every momentum-space exact-diagonalization copy (`exact_gs_energy`, five files), the real-space DMRG Hamiltonian (`build_H` in `dmrg_reference.jl/phase_scan_dmrg.jl`), and the real-space exact-diagonalization reference (`ed_reference`, same two files).

3.1 Momentum-space ED vs. real-space ED

Independent sparse exact diagonalizations in the two bases (`exact_gs_energy` vs. `ed_reference`), ground-state energy per (L, V) :

L	V	E_{momentum}	E_{real}	$ \Delta E $
4	0.5	-2.5894541729	-2.5894541729	4×10^{-15}
4	1.0	-2.3722813233	-2.3722813233	6×10^{-15}
4	2.0	-2.0000000000	-2.0000000000	1×10^{-14}
4	3.0	-1.7015621187	-1.7015621187	6×10^{-15}
8	0.5	-4.6769911638	-4.6769911638	5×10^{-15}
8	1.0	-4.1739887103	-4.1739887103	2×10^{-15}
8	2.0	-3.3021868179	-3.3021868179	4×10^{-15}
8	3.0	-2.6070908186	-2.6070908186	7×10^{-15}
12	0.5	-6.8859094308	-6.8859094308	1×10^{-14}
12	1.0	-6.1145448817	-6.1145448817	1×10^{-14}
12	2.0	-4.7747818349	-4.7747818349	4×10^{-15}
12	3.0	-3.7097076448	-3.7097076448	2×10^{-14}

Table 1: Ground-state energy match between the two exact-diagonalization constructions, machine precision throughout.

3.2 VMC local-energy estimator vs. ED (`test_paired_eloc.jl`)

For a trial (non-optimized) grouped-ansatz state, enumerated exactly over the full $N = L/2$ sector at $L = 8$ ($D = 70$ basis states):

$$\begin{aligned}
 V = 0.0: \quad & \langle \psi | H | \psi \rangle / \langle \psi | \psi \rangle = -5.0962372381, \quad \sum_s p(s) E_{\text{loc}}(s) = -5.0962372381 \quad (|\Delta| = 0) \\
 V = 0.7: \quad & \langle \psi | H | \psi \rangle / \langle \psi | \psi \rangle = -2.8241408216, \quad \sum_s p(s) E_{\text{loc}}(s) = -2.8241408216 \quad (|\Delta| = 3 \times 10^{-15})
 \end{aligned}$$

confirming `get_E_loc` — the estimator actually sampled during VMC/SR optimization — reproduces the exact variational energy of the same state to machine precision, i.e. the interaction-coefficient fix was applied consistently to the sampler itself, not only to the ED cross-checks.

3.3 Full DMRG optimizer vs. ED

Running the actual two-site DMRG optimizer (not just its ED reference) at $L = 8$, $V = 1.0$, $\chi = 16$:

$$E_{\text{DMRG}} = -4.17398871, \quad E_{\text{ED}} = -4.17398871, \quad \text{gap} = 1.3 \times 10^{-9}.$$

4 Convention history

An earlier version of every file listed above used a real-space coefficient of $2V n_i n_{i+1}$ (equivalently a momentum-space vertex of $2V \cos q$) under the *same* variable name V . This was originally *discovered* — not designed — by comparing the full eigenvalue spectrum (not just the ground state, which can match coincidentally at small L) between the momentum-space Hamiltonian and several candidate real-space Hamiltonians, and finding that a factor of 2 was needed for the two representations to agree. That factor-of-2 convention was traceable to how the momentum-space Hamiltonian was originally written down (inherited from an earlier, pre-grouped-ansatz version of this codebase) and was never an intentional physical choice.

The practical consequence: under the old convention, the LL→CDW transition (Eq. 13, $\Delta = 1$) occurred at $V_{\text{target}} = t$, not $V_{\text{target}} = 2t$ as the standard “ $V = 2t$ ” phrasing for the t – V model implies. Every file was then changed together (see §1.3) to drop the extra factor of 2, verified via all the checks in this section, restoring the standard convention $\Delta = V/(2t)$, $V_c = 2t$.